Juncai He: Curriculum Vitae

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Research Interest

- Theory and Algorithms of Deep Neural Networks and Operators, Stochastic Optimization.
- Numerical Analysis, Finite Element Methods, Multigrid Methods.

Academic Experience

- **Assistant Professor** February 2025 - present
  Yau Mathematical Sciences Center, Tsinghua University, Beijing, China
- **Research Scientist** July 2022 - January 2025
  Applied Mathematics and Computational Sciences, KAUST, Saudi Arabia
- **R.H. Bing Instructor Fellow** August 2020 - July 2022
  Department of Mathematics, The University of Texas at Austin, Austin, TX, USA
- **Postdoctoral Scholar** August 2019 - July 2020
  Department of Mathematics, The Pennsylvania State University, University Park, PA, USA

Education

- **Ph.D.**, Computational Mathematics, Peking University, 2014-2019
  Advisors: Prof. Jinchao Xu and Prof. Jun Hu
  Thesis: Finite Element Methods and Deep Neural Networks
- **Visiting Ph.D. Research Scholar**, Center for Computational Mathematics and Application, Department of Mathematics, The Pennsylvania State University, Feb. 2016 - Jul. 2016, Oct. 2017 - Mar. 2018 and Mar. 2019 - May 2019
- **B.S.**, Mathematics and Applied Mathematics, Sichuan University, 2010-2014

Honor and awards

- 2020-2022, R. H. Bing Fellowship, UT Austin.
- 2016-2019, The Elite Program of Computational and Applied Mathematics for PHD Candidates of Peking University
- 2017-2019, Ph.D. President Scholarship, Peking University
- 2015, Graduate academic scholarship, Peking University
- 2014, First Prize, Outstanding undergraduate thesis, Sichuan University
- 2011-2014, Excellent undergraduate student scholarship, Sichuan University

The Workshops/ Minisymposium Organized

- Minisymposium on "Analysis, Algorithms, and Applications of Neural Networks" (Co-organizer with Jinchao Xu and Xinliang Liu) and Minisymposium on "Understanding the Learning of Deep Networks: Expressivity, Optimization, and Generalization" (Co-organizer with Fenglei Fan and Shijun Zhang) at The 14th AIMS Conference, December 16-20, 2024, Abu Dhabi, UAE.
- CUHK-Shenzhen and CUHK Joint Summer School on Scientific Computing and Machine Learning (Co-Chair of the Organizing Committee), July 8-18, 2024, Chinese University of Hong Kong, Shenzhen, China.
- KAUST Research Conference on Scientific Computing and Machine Learning (Organizing Committee), November 14-18, 2022, KAUST, KSA.
- Minisymposium on “Multigrid and Multilayer Methods” (Co-organizer with Prof. Jinchao Xu and Dr. Xinliang Liu) in International Multigrid Conference , August 22-26, 2022, Lugano, Switzerland
- Workshop on Mathematical Machine Learning and Application (Organizing Committee), December 14-16, 2020, Penn State University, USA
- Minisymposium on “Multigrid and Machine Learning” (Co-organizer with Prof. Zuowei Shen and Prof. Jinchao Xu) in International Multigrid Conference , August 11-16, 2019, Kunming, China
- 4th PKU Workshop on Numerical Methods for PDEs (Organizing Committee), October 30-31, 2018, Peking University, China
- The First PKU Elite PHD Candidates Workshop on Computational Mathematics and 4th Beijing Graduate Students Workshop on Computational Mathematics (Chair of the Organizing Committee), September 9-12, 2018, Peking University, China

Invited Conference Talks

1. Conference on "Frontiers in Multiscale Computations", July 6 - July 10, **Institut Mittag-Leffler (IML)**, Sweden, 2026.
2. Invited Talks at The Tenth Triennial International Congress of Chinese Mathematicians (**ICCM 2025**), January 2-8, Shanghai, China, 2026.
3. Workshop on "Constrained Dynamics, Stochastic Numerical Methods and the Modeling of Complex Systems", May 26-31, **Mathematisches Forschungsinstitut Oberwolfach (MFO)**, Germany, 2024.
4. The Twelfth International Conference on Learning Representations (ICLR, poster), May 4-11, Vienna, Austria, 2024.
5. PhotonIcs and Electromagnetics Research Symposium (PIERS), April 21-25, Chengdu, China, 2024.
6. KAUST Rising Stars in AI Symposium, February 19th, KAUST, Saudi Arabia, 2024.
7. CBMS Conference: Deep Learning and Numerical PDEs, June 19-23, Morgan State University, USA, 2023.
8. SIAM Conference on Computational Science and Engineering (CSE23), February 26 - March 23, Amsterdam, Netherlands, 2023.
9. KAUST Research Conference on Scientific Computing and Machine Learning, November 14-18, KAUST, Saudi Arabia, 2022.

10. International Multigrid Conference (IMG2022), August 22-26, University of Lugano, Lugano, Switzerland, 2022.
11. The Finite Element Circus Fall 2021, Penn State University, University Park, USA, Nov. 2021.
12. IMA Workshop on Mathematical Foundation and Applications of Deep Learning, Purdue University, West Lafayette (Online), USA, Aug. 2021.
13. The First Young Scholar Forum, Peking University Chongqing Research Institute of Big Data, Chongqing (Online, invited talk), China, Jul. 2021.
14. Workshop on Mathematical Machine Learning and Application, Penn State University, University Park (Online, invited talk), USA, Dec. 2020.
15. Workshop on Computation and Applications of PDEs Based on Machine Learning, Jilin University, Changchun (Online, invited talk), China, Jul. 2020.
16. "Advances in Multilevel Methods: from PDEs to Data Intensive Studies" and "Multigrid and Machine Learning", Minisymposiums in International Multigrid Conference, Kunming, China, Aug. 2019.
17. 16th Annual Meeting of the China Society for Industrial and Applied Mathematics, Chengdu, China, Sept. 2018.
18. The First PKU Elite PHD Candidates Workshop on Computational Mathematics and 4th Beijing Graduate Students Workshop on Computational Mathematics, Peking University, Beijing, China, Sept. 2018.
19. Workshop on Numerical Methods for PDEs, Peking University, Beijing, China, Jul. 2017.

1 Invited Seminar and Colloquium Talks

1. School of Mathematics, April 24th, Sichuan University, China, 2024.
2. Yau Mathematical Sciences Center, April 11th, Tsinghua University, China, 2024.
3. CMAI: Artificial Intelligence Colloquium Series, August 4th, The Chinese Hong Kong University, 2023.
4. Applied and Computational Mathematics Seminar, UC Irvine, Irvine (Online), USA, Jan. 2022.
5. Data Science Seminar, Shanghai Jiao Tong University, Shanghai (Online), China, Mar. 2020.

Submitted and Preprints

1. **J. He** and Z. Tian. Sharp Sobolev Sandwich and Approximation Rates of Radon-Domain L^p Ridge Integral Spaces for ReLU^k Networks. ArXiv:2606.24795, 2026.
2. S. Chen, **J. He** and X.-C. Tai. Neural Flow Operators can Approximate any Operator: Abstract Frameworks and Universal Approcimations. ArXiv:2605.22557, 2026.
3. **J. He**, X. Liu and Z. Tian. Divergence-free Linearized Neural Networks: Integral Representation and Optimal Approximation Rates. ArXiv:2603.28638, 2026.

4. Y. Yang and **J. He**#. Deep Neural Networks with General Activations: Super-Convergence in Sobolev Norms. ArXiv:2508.05141, 2025.

5. H. Wu, Y. Gao, R. Shu, K. Wang, R. Gou, C. Wu, X. Liu, **J. He**, S. Cao, J. Fang, X. Shi, F. Tao, Q. Song, S. Ji, Y. Xiang, Y. Sun, J. Li, F. Xu, H. Dong, H. Wang, F. Zhang, P. Zhao, X. Wu, Q. Wen, D. Chen, and X. Huang. Advanced Long-term Earth System Forecasting by Learning the Small-scale Nature. ArXiv:2505.19432, 2025.

6. **J. He**, T. Mao and J. Xu. Expressivity and Approximation Properties of Deep Neural Networks with ReLU^k Activation. ArXiv:2312.16483, 2023.

Publications

1. **J. He** and J. Xu. Deep Neural Networks and Finite Elements of Any Order on Arbitrary Dimensions. To appear in *Mathematical Models and Methods in Applied Sciences*, 37(1), 2027. [ArXiv:2312.14276].

2. **J. He**, X. Liu and J. Xu. Self-composing neural operators for high-frequency and multiscale PDE surrogates. *Journal of Computational Physics*. 115189, 2026. DOI: 10.1016/j.jcp.2026.115189 [ArXiv:2508.20650].

3. **J. He**, L. Liu and R. Tsai. Data-Induced Multiscale Losses and Efficient Multirate Gradient Descent Schemes. *Journal of Intelligent Algorithms and Scientific Computing*. 1(1): 44–87, 2026. DOI: 10.4208/jiasc.2026-91-1 [ArXiv:2402.03021].

4. Z. Yang, Z. Tian, Y. Jia, T. Zhang, J. Zheng, H. Wang, Y. Su, **J. He**, L. Liu, Y. Lan. Cross-Chirality Generalization by Axial Vectors for Hetero-Chiral Protein-Peptide Interaction Design. ArXiv:2602.20176, 2026. Accepted by *ICML 2026*.

5. G. Bao, Y. Zhao, **J. He**, Y. Zhang. Glimpse: Enabling White-Box Methods to Use Proprietary Models for Zero-Shot LLM-Generated Text Detection. Accepted for *ICLR 2025*. [ArXiv:2412.11506]

6. **J. He**. On the Optimal Expressive Power of ReLU DNNs and Its Application in Approximation with Kolmogorov Superposition Theorem. *IEEE Transactions on Neural Networks and Learning Systems*, 2024. DOI: 10.1109/TNNLS.2024.3514126 [ArXiv:2308.05509].

7. J. Liang, Z. Cai, J. Zhu, H. Huang, K. Zong, B. An, M. Alharthi, **J. He**, L. Zhang, H. Li, B. Wang and J. Xu. Alignment at Pre-training! Towards Native Alignment for Arabic LLMs. *NeurIPS 2024* [ArXiv:2412.03253].

8. J. Zhu, H. Huang, Z. Lin, J. Liang, Z. Tang, K. Almubarak, M. Alharthi, B. An, **J. He**, X. Wu, F. Yu, J. Chen, Z. Ma, Y. Du, Y. Hu, H. Zhang, E. Alghamdi, L. Zhang, R. Sun, H. Li, J. Xu, B. Wang. Second Language (Arabic) Acquisition of LLMs via Progressive Vocabulary Expansion. Accepted by *ACL 2025*.

9. Y. Yang and **J. He**#: Deeper or Wider: A Perspective from Optimal Generalization Error with Sobolev Loss. ICML 2024, [ArXiv:2402.00152].

10. H. Huang, F. Yu, J. Zhu, X. Sun, H. Cheng, D. Song, Z. Chen, M. Alharthi, B. An, **J. He**, Z. Liu, Z. Zhang, J. Chen, J. Li, B. Wang, L. Zhang, R. Sun, X. Wan, H. Li, J. Xu. AceGPT, Localizing Large Language Models in Arabic. *NAACL 2024*. [ArXiv:2309.12053].

11. **J. He**, X. Liu and J. Xu. MgNO: Efficient Parameterization of Linear Operators via Multigrid. *ICLR 2024*, [ArXiv:2310.19809], [ResearchGate].

12. L. Liu, **J. He** and R. Tsai. Linear Regression on Manifold Structured Data: The Impact of Extrinsic Geometry on Solutions. *Topological, Algebraic and Geometric Learning Workshops at ICML2023*. PMLR 221:557-576, [Published PDF] 2023. [ArXiv:2307.02478].

13. J. Zhu*, **J. He*** and Q. Huang. An Enhanced V-cycle MgNet Model for Operator Learning in Numerical Partial Differential Equations. *Computational Geosciences* 2023. https://doi.org/10.1007/s10596-023-10211-8 [ArXiv:2302.00938].

14. J. Zhu, **J. He**, L. Zhang and J. Xu. FV-MgNet: Fully Connected V-cycle MgNet for Interpretable Time Series Forecasting. *Journal of Computational Science*. 69: 102005, 2023. https:doi.org10.1016j.jocs.2023.102005 [ArXiv:2302.00962].

15. **J. He**, J. Xu, L. Zhang and J. Zhu. An Interpretive Constrained Linear Model for ResNet and MgNet. *Neural Networks*. 162: 384-392, 2023. https://doi.org/10.1016/j.neunet.2023.03.011 [ArXiv:2112.07441]

16. **J. He**, R. Tsai and R. Ward. Side Effects of Learning from Low-dimensional Data Embedded in a Euclidean Space. *Research in the Mathematical Sciences*. 10(13), 2023. https:doi.org10.1007s40687-023-00378-y. [ArXiv:2203.00614]

17. **J. He**, L. Li and J. Xu. ReLU Deep Neural Networks from the Perspective of Hierarchical Basis. *Computers & Mathematics with Applications*. 120: 105-114, 2022. https:doi.org10.1016j.camwa.2022.06.006 [ArXiv:2105.04156].

18. **J. He**, L. Li and J. Xu. Approximation Properties of Deep ReLU CNNs. *Research in the Mathematical Sciences.* 9(38), 2022. https:doi.org10.1007s40687-022-00336-0. [ArXiv:2109.00190].

19. Q. Chen, W. Hao and **J. He**. Power Series Expansion Neural Network. *Journal of Computational Science*. 59, 2022. https:doi.org10.1016j.jocs.2021.101552. [ArXiv:2102.13221].

20. Q. Chen, W. Hao and **J. He**. A Weight Initialization Based on the Linear Product Structure for Neural Networks. *Applied Mathematics and Computation*. 415, 2022. https://doi.org/10.1016/j.amc.2021.126722 [ArXiv: 2109.00125].

21. **J. He**, X. Jia, J. Xu, L. Zhang and L. Zhao. Make ℓ_1 Regularization Effective in Training Sparse CNN. *Computational Optimization and Applications*. 77: 163–182, 2020. https://doi.org/10.1007/s10589-020-00202-1

22. **J. He**, L. Li, J. Xu, and C. Zheng. ReLU Deep Neural Networks and Linear Finite Elements. *Journal of Computational Mathematics*. 38(3): 502–527, 2020. doi:10.4208/jcm.1810-m2018-0096. [ESI Highly Cited Paper in Mathematics and WOS Core Collection],[**Google Scholar Citation: 520+**].

23. **J. He**, K. Hu and J. Xu. Generalized Gaffney Inequality and Discrete Compactness for Discrete Differential Forms. *Numerische Mathematik*. 143, 781–795, 2019 . https://doi.org/10.1007/s00211-019-01076-0.

24. **J. He** and J. Xu. MgNet: A Unified Framework of Multigrid and Convolutional Neural Network. *Science China Mathematics*. 62(7): 1331–1354, 2019. https://doi.org/10.1007/s11425-019-9547-2. [ESI Highly Cited Paper in Mathematics and WOS Core Collection] [**Google Scholar Citation: 180+**].

#: corresponding author; *: equal contributions